Mangarapu Sree Arnav

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PROFESSIONAL SUMMARY

Computer Science undergraduate with experience in Computer Vision, Full-Stack Development, and AI/ML. Interested in Machine Learning, Infrastructure, and Cybersecurity, with hands-on experience building production-oriented software.

EDUCATION

Keshav Memorial Institute of Technology — B.Tech, Computer Science and Engineering Aug 2024 – 2028

PROJECTS

Helivision — Smart Helmet for Partially Visually Impaired Riders 2025

- Developed an embedded computer vision pipeline on Raspberry Pi using TensorFlow Lite for real-time vehicle detection and directional assistance.
- Integrated Raspberry Pi, I2C OLED displays, and optimized TensorFlow Lite models for low-latency inference on resource-constrained hardware.
- Improved system stability by migrating the camera pipeline to `picamera2`, resolving OS compatibility issues and restoring reliable real-time inference.

StreamWatch — Real-Time Database Monitoring and Observability Platform 2025

- Built a full-stack observability platform using React, Node.js, Express, Prometheus, and Grafana for real-time monitoring of database and infrastructure health.
- Configured Prometheus scrape pipelines and MongoDB Exporters on Linux systems to collect, aggregate, and visualize operational metrics.
- Diagnosed and resolved distributed service communication failures, improving monitoring reliability across the observability stack.

SRDC — Semantics-based Ransomware Detection using GPT-2 2026

- Built a cybersecurity language modeling pipeline by converting Windows API, registry, file-system, and behavioural logs into semantic text for transformer-based learning.
- Implemented data preprocessing and feature engineering workflows using Python for behaviour-based malware analysis.
- Fine-tuned GPT-2 using Hugging Face Transformers to generate semantic representations for ransomware classification.

TECHNICAL SKILLS

Languages:	Python, Java, C++, JavaScript, SQL
Web & Backend:	Node.js, Express, React, FastAPI
AI/ML:	CNN, ANN, TensorFlow
Databases:	MongoDB
Developer Tools:	Git, GitHub, Docker, Linux
Monitoring:	Prometheus, Grafana
Embedded Systems:	Raspberry Pi, Picamera2
Core CS:	Data Structures, OOP, Operating Systems, DBMS, Computer Networks

ACHIEVEMENTS

- Published Indian Patent Application (#202541100066)
- 2nd Prize — PRAKALP 2025 Innovation Competition

CERTIFICATIONS

OCI AI Foundations Associate (2025) • Anthropic: Introduction to Model Context Protocol • Anthropic: Model Context Protocol – Advanced Topics